USB TO ETHERCAT ADAPTER BOARD

The domain of the Project:

COURSE NAME

PCB DESIGNING

Team Mentors (and their designation):

PARAMESH KUMAR

Team Members:

Mr. A. PHANI DURGA PRASAD

Period of the project

AUGUST 2026 to SEPTEMBER 2026

Declaration

The project titled "USB TO ETHERCAT ADAPTER BOARD" has been mentored by Paramesh Kumar Sir organised by SURE Trust, from August 2026 to September 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Members: A. Phani Durga Prasad

Mr. A Phani durga prasad

Mentor's Name: Paramesh Kumar

Designation—SURE Trust

Prof. Radhakumari

Executive Director & Founder

SURE Trust

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Executive Summary

The USB-to-Etherncat Adapter Board is a two-layer printed circuit board (PCB) designed to provide an interface between USB connectivity and Ethernet communication. The project was developed as part of the PCB Design Internship using KiCad. The design integrates a microcontroller-based control section with USB, Ethernet/RJ45 interfaces, power regulation, EEPROM, and supporting electronic components on a single PCB.

The project involved the complete PCB design process, beginning with the preparation and review of the circuit schematic. The schematic includes an STM32 microcontroller, USB interface, two RJ45 interfaces, an I²C EEPROM section, power-supply circuitry for converting 5 V to 3.3 V, clock circuitry, filtering/decoupling components, and supporting connectors and components. Appropriate component footprints were selected and assigned before transferring the design to the PCB layout.

The PCB layout was developed as a two-layer board with consideration given to component placement, routing, board dimensions, clearances, mounting holes, and electrical connections. The completed design was reviewed using KiCad's PCB layout and 3D-viewing features. Both the front and back sides of the PCB were examined to verify the placement of components and overall physical arrangement.

The project provided practical experience in schematic design, component and footprint selection, PCB placement, routing, two-layer PCB development, and design verification using KiCad. The completed PCB design demonstrates the application of PCB design principles to an electronics communication interface and provides a foundation for further manufacturing, assembly, and hardware testing.

Introduction

Printed Circuit Boards (PCBs) are an essential part of modern electronic systems because they provide electrical connections and mechanical support for electronic components. PCB design involves converting an electronic circuit into a physical board by designing the schematic, selecting suitable components and footprints, placing components, routing electrical connections, and verifying the final layout.

The project presented in this report is a USB-to-Ethercat Adapter Board designed using KiCad. The board integrates an STM32 microcontroller with USB and Ethernet/RJ45 interface sections along with supporting circuits. The schematic also includes an I^2C EEPROM section, power-supply circuitry, clock circuits, filtering and decoupling capacitors, connectors, and other supporting components required for the board design.

The main goal of the project is to develop a complete two-layer PCB design from the circuit schematic. The design process includes schematic development, component and footprint selection, PCB component placement, routing of electrical connections, board-outline definition, and 3D visualization of the completed board. The PCB layout was designed to provide proper electrical connectivity and an organized physical arrangement of the components.

Project Objectives

The main objective of this project is to design and develop a two-layer USBto-Ethercat Adapter Board using KiCad . The project focuses on applying a complete PCB design workflow, from circuit schematic development to the final PCB layout and 3D visualization.

The specific objectives of the project are:

1. To understand the circuit requirements and develop the schematic for the USB-to-Ethercat Adapter Board.
2. To design and organize the circuit schematic using KiCad Schematic Editor.
3. To select appropriate electronic components according to the circuit requirements and component specifications.
4. To assign suitable PCB footprints to the selected components and verify their pin connections and package types.
5. To develop a two-layer PCB layout based on the completed schematic.
6. To place components efficiently on the PCB while considering connectivity, board space, and practical PCB design requirements.
7. To route the electrical connections between the components while maintaining appropriate track widths and clearances.
8. To design the PCB with the required interfaces and supporting circuits, including the USB interface, RJ45/Etherncat interface, STM32 microcontroller section, EEPROM, power-supply circuitry, clock circuitry, and supporting components shown in the schematic.
9. To verify the physical arrangement of the PCB using KiCad's 3D Viewer and examine both the front and back sides of the board.
10. To gain practical knowledge of industry-oriented PCB design and improve skills in schematic design, footprint selection, component placement, routing, and PCB layout development.

Methodology

The PCB design was developed using a systematic design workflow in KiCad. The methodology followed the stages of schematic development, component selection, footprint assignment, PCB layout development, component placement, routing, and 3D visualization.

The major steps followed during the project were:

→ Circuit Requirements
→ Schematic Design
→ Component Selection
→ Footprint Assignment
→ PCB Layout
→Component Placement
→ Routing
→ Design Verification
→ 3D Visualization

Tools and Software Used

The tools and software were used during the project:

KiCad - Schematic and PCB design

KiCad Schematic Editor – Circuit schematic development

KiCad PCB Editor – Component placement and PCB routing

KiCad 3D Viewer – Visualization of the completed PCB

Electronic component datasheets – Component/package reference

Project Architecture

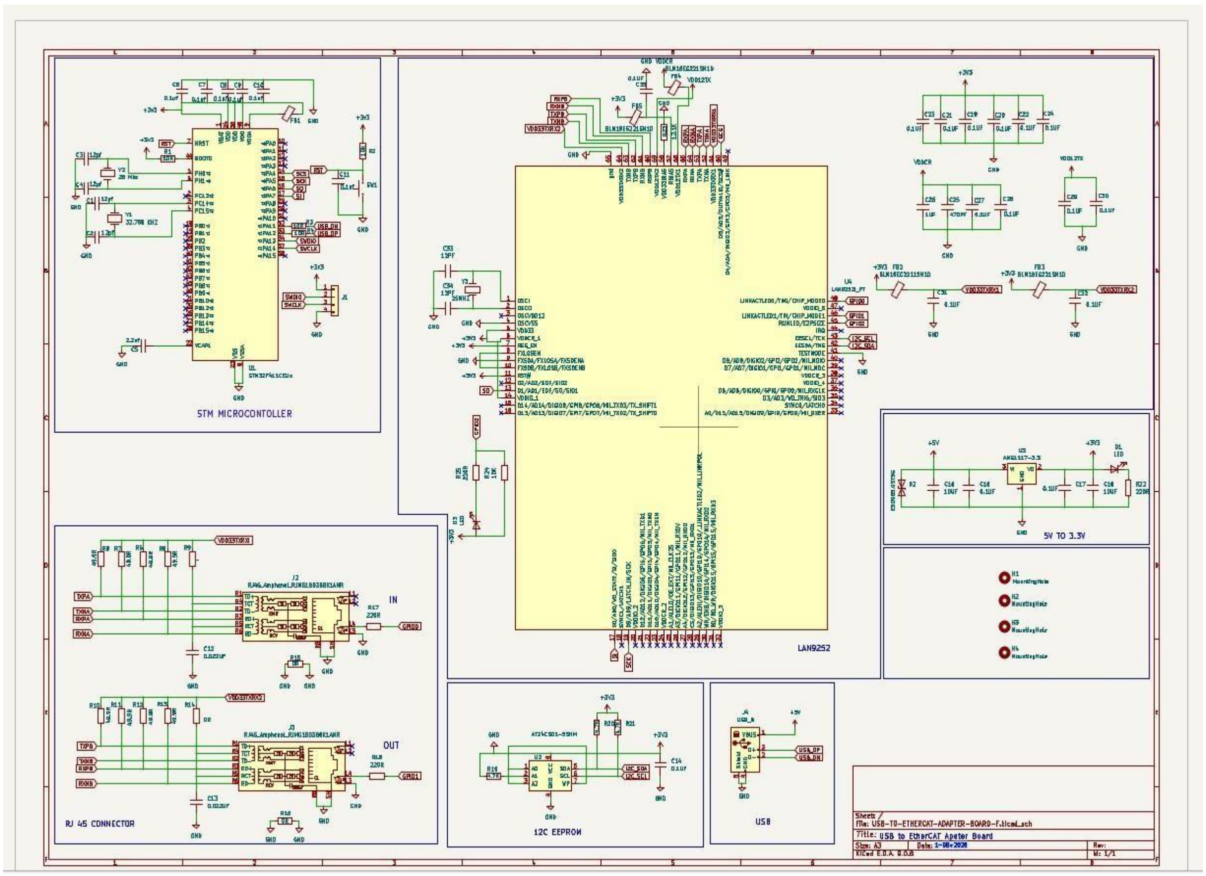
The USB-to-Ethercat Adapter Board consists of several functional sections integrated on a single PCB.

The major sections visible in the project schematic include:

1. STM32 microcontroller section
2. USB interface section
3. Ethernet/RJ45 interface section
4. EEPROM section
5. Power-supply section
6. Clock/oscillator section
7. Filtering and decoupling section
8. Connectors and supporting components

Final Project Working

Schematic Design



The circuit schematic was developed using the KiCad Schematic Editor. The required electronic components were placed and electrically connected according to the project requirements.

The schematic includes the microcontroller, USB interface, Ethernet/RJ45 interfaces, EEPROM, power circuitry, clock circuitry, capacitors, resistors, connectors, and other supporting components.

Explanation

The schematic represents the complete electrical design of the project. It shows the interconnections between the microcontroller, communication interfaces, power circuits, memory section, clock circuit, and supporting components.

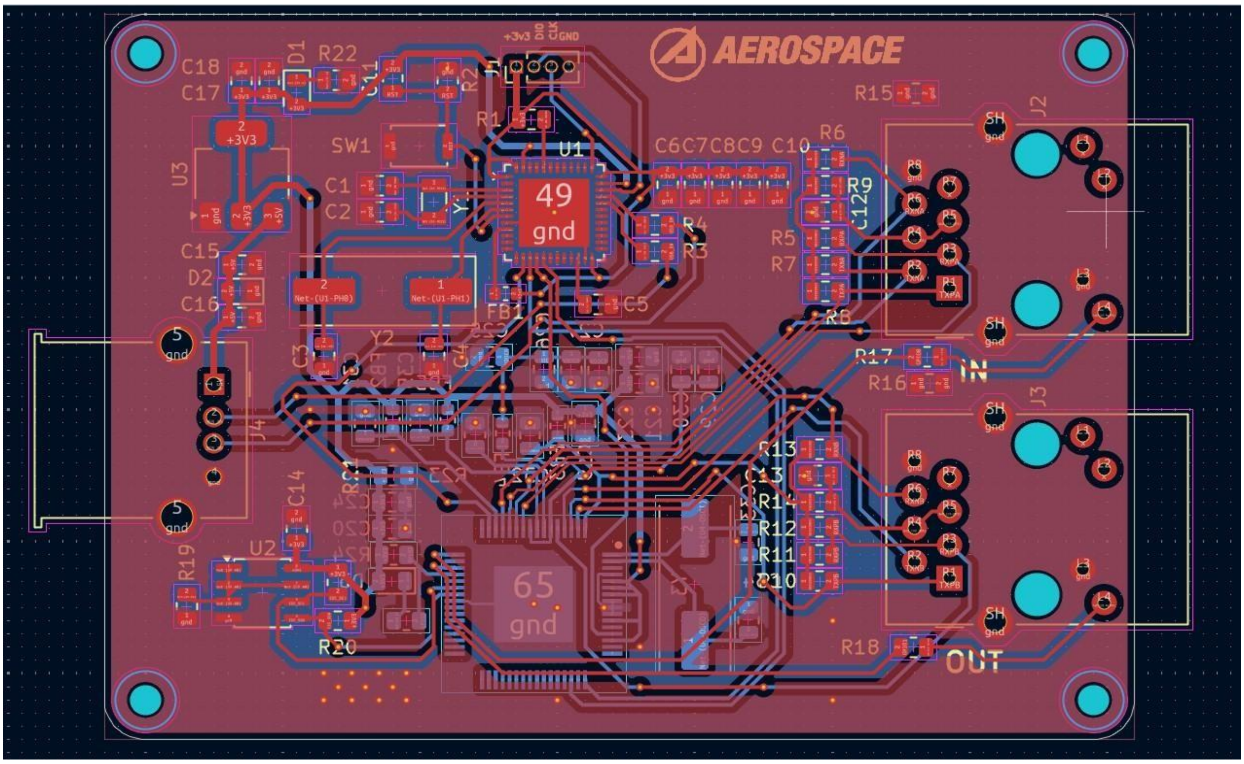
Component and Footprint Selection

After completing the schematic, suitable PCB footprints were assigned to the components. Footprints were selected according to the physical package and pin configuration of each component.

The footprint assignment ensures that the schematic symbols are correctly represented on the physical PCB layout and that the component pads correspond correctly to the component pins.

PCB Layout Design

The schematic was transferred to the KiCad PCB Editor to create the physical PCB layout. The board outline was defined and the components were positioned according to their electrical connections and physical requirements. Component placement was organized to provide a practical and structured PCB layout



Explanation

The PCB layout shows the physical arrangement of the components and the routed electrical connections. The layout was developed as a two-layer PCB with components and tracks arranged according to the design requirements.

PCB Routing

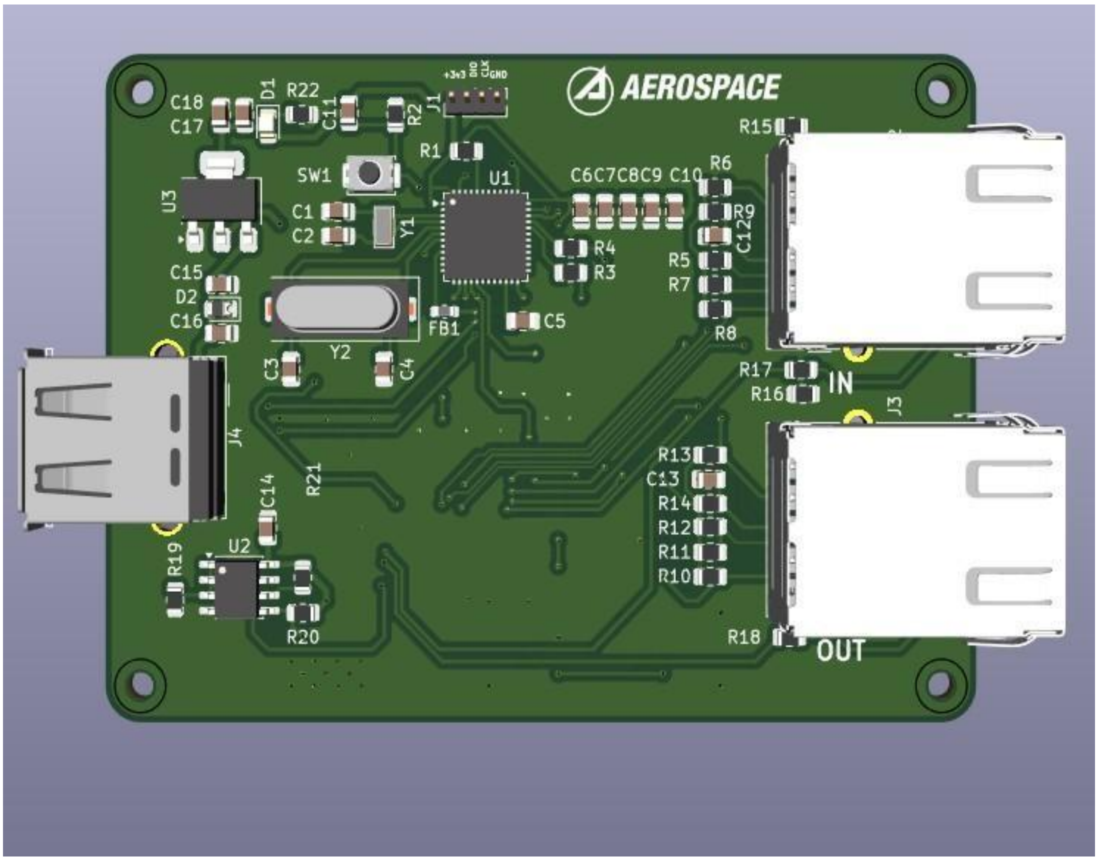
After component placement, the electrical connections were routed on the PCB. Routing was performed by connecting the required component pads while maintaining suitable track paths and clearances.

The routing process converts the electrical connections in the schematic into physical copper tracks on the PCB.

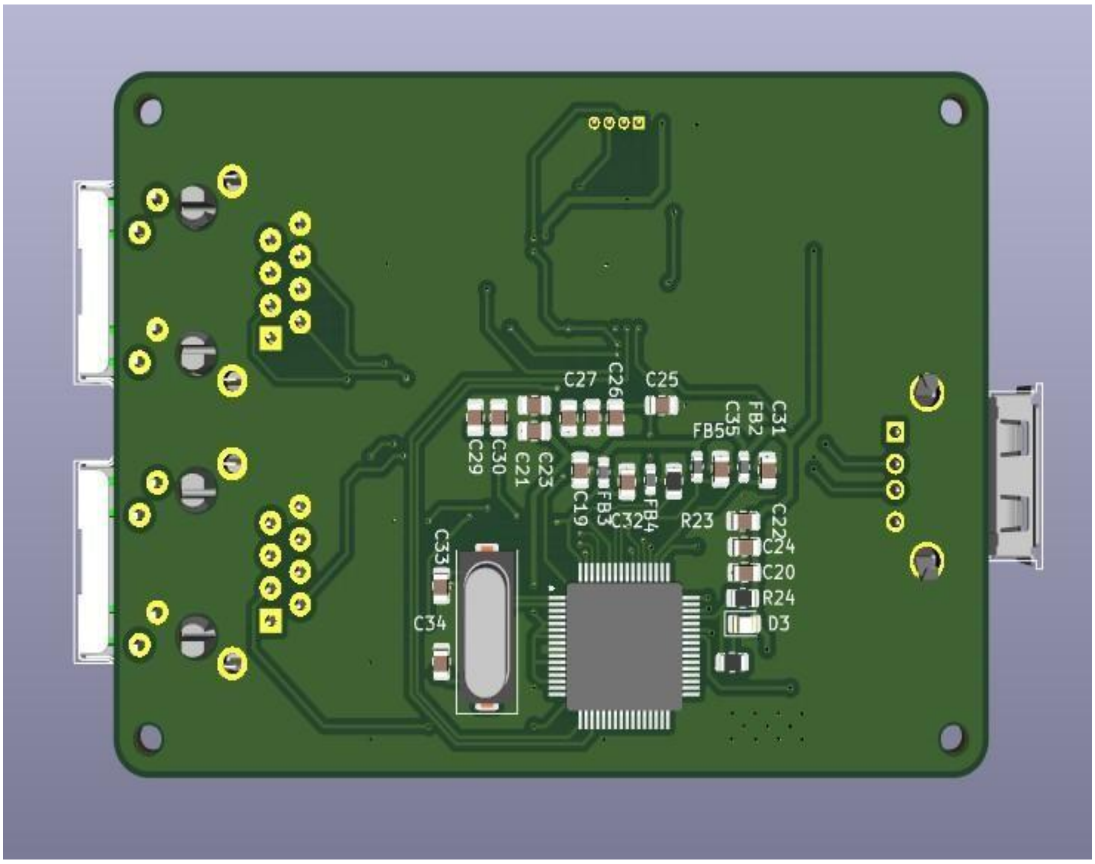
3D PCB Visualization

The completed PCB layout was examined using the KiCad 3D Viewer. The 3D view helps visualize the physical arrangement of components and the overall appearance of the designed board.

Front-Side 3D View of the PCB



Back-Side 3D View of the PCB



Explanation

The front and back 3D views provide a visual representation of the completed PCB and help verify the component arrangement, board outline, connectors, and overall physical structure.

Results

The final PCB layout of the USB-to-Ethernet Adapter Board was completed using KiCad. The schematic was converted into a physical two-layer PCB layout with appropriate component placement and routing. The completed design was also visualized using the KiCad 3D Viewer.

The project provided practical experience in the complete PCB design workflow, from schematic development to PCB layout and 3D visualization.

Project GitHub Link

https://github.com/ayodhyaphanidurgaprasad-source/PCB-Projects.git

Technical Learning

Through this project and my PCB Design internship, I gained practical knowledge of the complete PCB design process using KiCad I learned how to understand circuit requirements and develop a schematic by placing electronic components and creating the required electrical connections.

I learned how to select suitable components and assign appropriate PCB footprints based on their package types and pin configurations. I also gained practical experience in transferring the schematic to the PCB Editor and developing a two-layer PCB layout.

During PCB layout development, I learned about component placement, board-outline creation, track routing, clearances, and organizing different functional sections of a PCB. I also learned how to use the KiCad 3D Viewer to visualize the completed PCB from different sides.

Working on the USB-to-Ethernet Adapter Board helped me understand how a schematic is converted into a physical PCB design and how different electronic sections can be integrated into a single board.

Overall Reflection

Overall, this project was a valuable hands-on learning experience. It helped me connect the theoretical concepts of electronics and PCB design with practical hardware development. I gained more confidence in working with PCB design software and developed a better understanding of the workflow followed in PCB design projects.

The project also motivated me to continue improving my PCB design skills and gain further experience with advanced PCB layout, design verification, manufacturing processes, and real-world hardware testing.

Conclusion

The project successfully demonstrated the complete PCB design workflow using KiCad. The circuit schematic was developed, appropriate footprints were assigned, components were placed, PCB routing was completed, and the design was verified using ERC and DRC checks. The final PCB was also reviewed using the 3D viewer.

This project provided practical experience in PCB design and helped improve my technical knowledge of electronic hardware development. The skills gained through this project can be further developed through advanced PCB design and real-world hardware projects

Future Scope

The future scope of the project includes manufacturing and assembling the designed PCB, testing the physical board, and validating its electrical performance. Further improvements can include optimizing the PCB layout, improving signal integrity and thermal management where required, and developing an enhanced version with additional features based on application requirements..